

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Solve/ Answer Any Two of the following.		12
A)	Derive an expression for Helmholtz's equation for the wave propagating in lossy dielectric.	01	6
B)	State Poynting's theorem. Also derive an expression for Poynting's theorem.	01	6
C)	A uniform plane wave propagating in a good conductor has $\mathbf{E} = e^{-\alpha z} \sin(10^8 t - \beta z) \mathbf{a}_y \text{ V/m}$	01	6
	If the medium is characterized by $\epsilon_r = 1$, $\mu_r = 20$ and $\sigma = 4 \text{ S/m}$, find α , β , and $\mathbf{H}$.		
Q.2	Solve/ Answer Any Two of the following.		12
A)	Explain in brief: a) Ground wave propagation, b) Space wave propagation, c) Sky wave propagation.	03	6
B)	Explain structure of atmosphere in brief.	03	6
C)	Calculate critical frequencies for F1, F2, and E-Layer for which the maximum ionic densities are $2.5 \times 10^6 \text{ cm}^{-3}$, $4 \times 10^6 \text{ cm}^{-3}$ and $2 \times 10^6 \text{ cm}^{-3}$ respectively.	03	6
Q. 3	Solve/ Answer Any Two of the following.		12
A)	Explain radiation mechanism for a) single wire antenna and b) two wire antenna.	02	6
B)	Define: a) radiation pattern, b) radiation power density, c) directivity and d) gain.	02	6
C)	Determine the radiation resistance of an antenna if it has a loss resistance of 20Ω , power gain of 50 and directivity of a) 70 and b) 60.	02	6
Q.4	Solve/ Answer Any Two of the following.		12
A)	Find $\mathbf{E}$ and $\mathbf{H}$ fields for Hertzian dipole antenna with uniform current distribution.	02	6
B)	What are dipole arrays? Explain a) Broadside array and b) Endfire array.	02,03	6
C)	Write a short note on Yagi – Uda antenna.	03	6
Q. 5	Solve/ Answer Any Two of the following.		12
A)	Explain travelling wave Antennas in detail.	03	6
B)	Explain in detail : Microstrip Antenna	03	6
C)	Explain Cassegrain feed parabolic reflector with suitable diagram.	03	6

*** End ***